Worked DSP project solution

The implementation is in `dsp_solution.py`. This notebook runs all experiments and displays measured evidence. Parameters and theoretical interpretation are in the accompanying solution document. The saved outputs below are from a completed run.

```
from run_solution import main
metrics = main()
```


```
{
  "time_frequency": [
    {
      "L": 256,
      "window_ms": 16.0,
      "bin_hz": 62.5,
      "hop_ms": 4.0,
      "transient_10percent_energy_width_ms": 8.0
    },
    {
      "L": 1024,
      "window_ms": 64.0,
      "bin_hz": 15.625,
      "hop_ms": 16.0,
      "transient_10percent_energy_width_ms": 32.0
    },
    {
      "L": 4096,
      "window_ms": 256.0,
      "bin_hz": 3.90625,
      "hop_ms": 64.0,
      "transient_10percent_energy_width_ms": 192.0
    }
  ],
  "windows": [
    {
      "window": "boxcar",
      "null_to_null_hz": 31.25,
      "peak_sidelobe_db": -13.263647577714902,
      "enbw_hz": 15.625
    },
    {
      "window": "hann",
      "null_to_null_hz": 62.5,
      "peak_sidelobe_db": -31.467626478524792,
      "enbw_hz": 23.4375
    },
    {
      "window": "hamming",
```

```
    "null_to_null_hz": 62.5,
    "peak_sidelobe_db": -42.6743663692628,
    "enbw_hz": 21.294152949245543
  },
  {
    "window": "blackman",
    "null_to_null_hz": 93.75,
    "peak_sidelobe_db": -58.10884368817948,
    "enbw_hz": 26.98058390022675
  }
],
"overlap": [
  {
    "overlap_percent": 0,
    "hop": 1024,
    "window": "boxcar",
    "frames": 65,
    "relative_error": 2.6735228456948584e-16,
    "min_denominator": 1.0,
    "max_denominator": 1.0,
    "elapsed_seconds": 0.04068229999393225
  },
  {
    "overlap_percent": 0,
    "hop": 1024,
    "window": "hann",
    "frames": 65,
    "expected_failure": "NOLA failure in retained samples",
    "elapsed_seconds": 0.0032158999238163233
  },
  {
    "overlap_percent": 25.0,
    "hop": 768,
    "window": "hann",
    "frames": 86,
    "relative_error": 3.7200893184236793e-16,
    "min_denominator": 0.0428932188134525,
    "max_denominator": 1.0,
    "elapsed_seconds": 0.004012899938970804
  },
  {
    "overlap_percent": 50.0,
    "hop": 512,
    "window": "hann",
    "frames": 128,
    "relative_error": 2.444013099020329e-16,
    "min_denominator": 0.5,
    "max_denominator": 1.0,
    "elapsed_seconds": 0.005510699935257435
  },
```

```
    {
      "overlap_percent": 75.0,
      "hop": 256,
      "window": "hann",
      "frames": 255,
      "relative_error": 2.0087174059560197e-16,
      "min_denominator": 1.4999999999999993,
      "max_denominator": 1.5000000000000007,
      "elapsed_seconds": 0.010108600021339953
    },
    {
      "overlap_percent": 87.5,
      "hop": 128,
      "window": "hann",
      "frames": 509,
      "relative_error": 1.9368621161930796e-16,
      "min_denominator": 2.999999999999999,
      "max_denominator": 3.0000000000000013,
      "elapsed_seconds": 0.020057399990037084
    }
  ],
  "ola": {
    "relative_error": 3.267306983317611e-16,
    "output_samples": 64128,
    "fft_length": 1024,
    "group_delay_samples": 64
  },
  "ola_block_sweep": [
    {
      "B": 128,
      "nfft": 256,
      "blocks": 500,
      "relative_error": 3.486251505589894e-16
    },
    {
      "B": 512,
      "nfft": 1024,
      "blocks": 125,
      "relative_error": 3.267306983317611e-16
    },
    {
      "B": 2048,
      "nfft": 4096,
      "blocks": 32,
      "relative_error": 3.4427220579912124e-16
    }
  ],
  "jammer": {
    "input_snr_db": -9.241890416521894,
    "output_snr_db": 17.349087100818082,
```

```
  "stationary_notch_snr_db": -8.481089473614823
},
"jammer_sweep": [
  {
    "L": 256,
    "width_hz": 30,
    "output_snr_db": -0.8024310924202096,
    "jammer_attenuation_db": 8.439462222975777,
    "clean_relative_error": 0.00024308965005550364
  },
  {
    "L": 256,
    "width_hz": 60,
    "output_snr_db": 4.843103849910849,
    "jammer_attenuation_db": 14.08499339328788,
    "clean_relative_error": 0.00037779775091329127
  },
  {
    "L": 256,
    "width_hz": 120,
    "output_snr_db": 12.481593042684294,
    "jammer_attenuation_db": 21.72345862836615,
    "clean_relative_error": 0.0005655553072143046
  },
  {
    "L": 1024,
    "width_hz": 30,
    "output_snr_db": 11.76507671056524,
    "jammer_attenuation_db": 21.007007364925418,
    "clean_relative_error": 0.0002810228251903903
  },
  {
    "L": 1024,
    "width_hz": 60,
    "output_snr_db": 17.349087100818082,
    "jammer_attenuation_db": 26.590936278608414,
    "clean_relative_error": 0.000404427480612134
  },
  {
    "L": 1024,
    "width_hz": 120,
    "output_snr_db": 19.878447725307534,
    "jammer_attenuation_db": 29.120208120571572,
    "clean_relative_error": 0.0005753947233051704
  },
  {
    "L": 4096,
    "width_hz": 30,
    "output_snr_db": 11.350648144222896,
    "jammer_attenuation_db": 20.59250594230059,
```

```
      "clean_relative_error": 0.00027841472515402395
    },
    {
      "L": 4096,
      "width_hz": 60,
      "output_snr_db": 17.0753710187401,
      "jammer_attenuation_db": 26.317142591547835,
      "clean_relative_error": 0.0004034867182518259
    },
    {
      "L": 4096,
      "width_hz": 120,
      "output_snr_db": 19.76949495780763,
      "jammer_attenuation_db": 29.01121795144676,
      "clean_relative_error": 0.0005751797356772257
    }
  ],
  "iq_wav_roundtrip_error": 2.3820580951013048e-08,
  "communications": {
    "original": {
      "evm_percent": 0.947661602826916,
      "ber": 0.0,
      "waveform_relative_error": 0.0
    },
    "coherent": {
      "evm_percent": 0.9476616028269153,
      "ber": 0.0,
      "waveform_relative_error": 2.3268009963511954e-16
    },
    "magnitude_only": {
      "evm_percent": 101.02894144350883,
      "ber": 0.4990138067061144,
      "waveform_relative_error": 1.0103187240492075
    },
    "frame_phase_reset": {
      "evm_percent": 127.67102277627777,
      "ber": 0.5226824457593688,
      "waveform_relative_error": 1.2816860544023
    }
  },
  "checks": "PASS: identity, expected NOLA failure, real and complex full
OLA, QPSK BER"
}
```

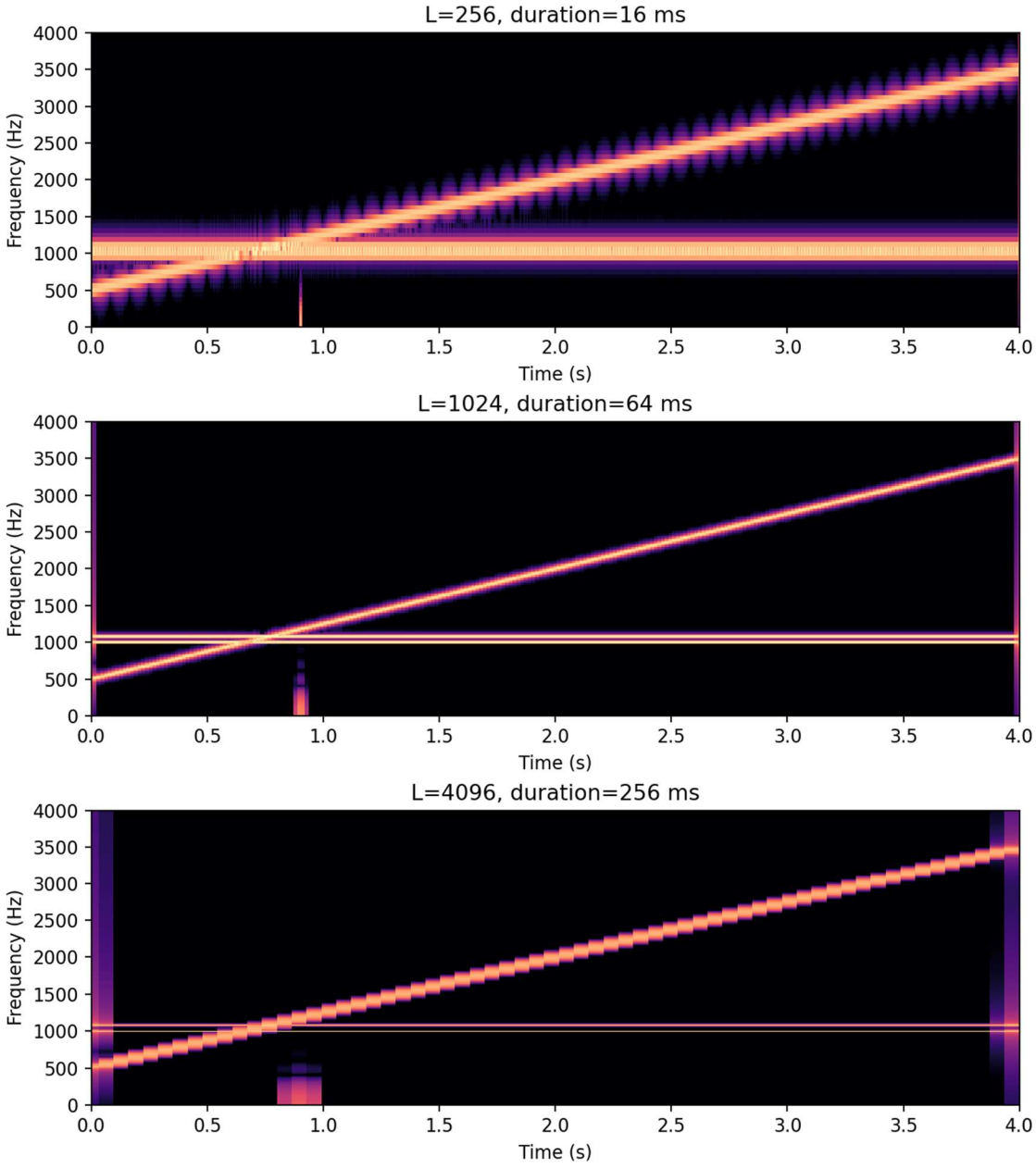

Time frequency tradeoff

```
print(metrics['time_frequency'])
```


```
[{'L': 256, 'window_ms': 16.0, 'bin_hz': 62.5, 'hop_ms': 4.0,
'transient_10percent_energy_width_ms': 8.0}, {'L': 1024, 'window_ms': 64.0,
```

```
'bin_hz': 15.625, 'hop_ms': 16.0, 'transient_10percent_energy_width_ms':
32.0}, {'L': 4096, 'window_ms': 256.0, 'bin_hz': 3.90625, 'hop_ms': 64.0,
'transient_10percent_energy_width_ms': 192.0}]
```

```
from IPython.display import display, Image
display(Image(filename="results/uncertainty.png"))
```



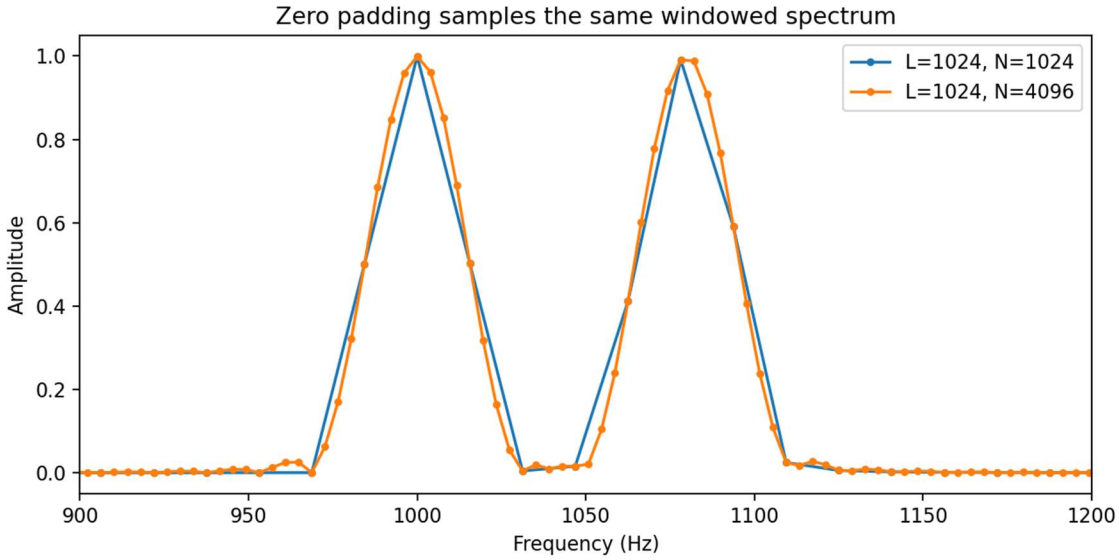
Zero padding

```
print(metrics['time_frequency'])
```

```
[{'L': 256, 'window_ms': 16.0, 'bin_hz': 62.5, 'hop_ms': 4.0,
'transient_10percent_energy_width_ms': 8.0}, {'L': 1024, 'window_ms': 64.0,
```

```
'bin_hz': 15.625, 'hop_ms': 16.0, 'transient_10percent_energy_width_ms':
32.0}, {'L': 4096, 'window_ms': 256.0, 'bin_hz': 3.90625, 'hop_ms': 64.0,
'transient_10percent_energy_width_ms': 192.0}]
```

```
from IPython.display import display, Image
display(Image(filename="results/zero_padding.png"))
```

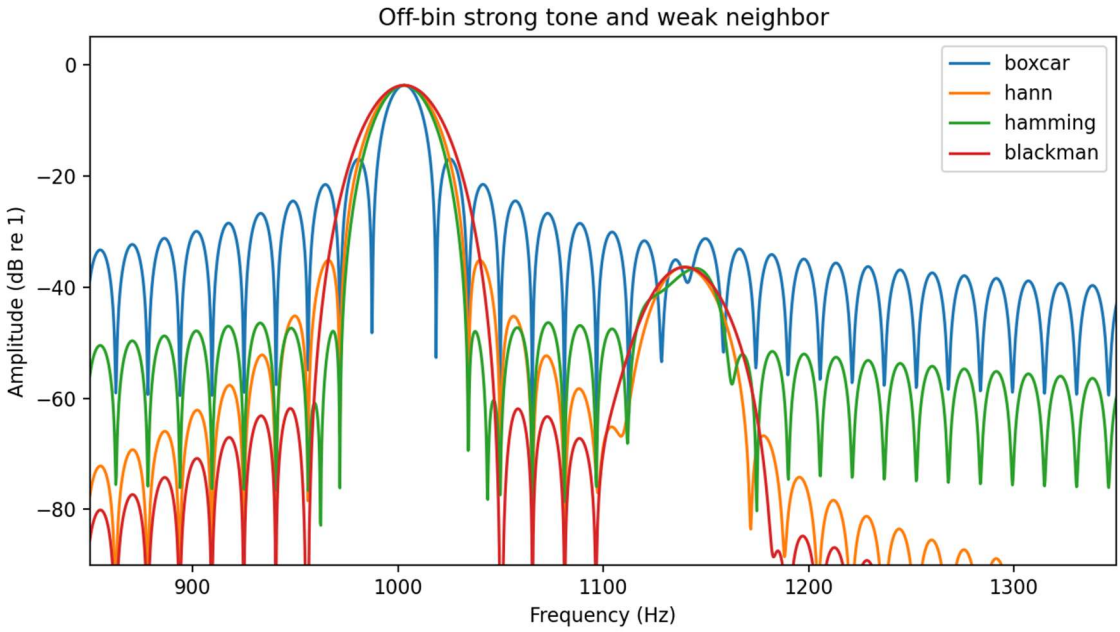


Windows and leakage

```
print(metrics['windows'])
```

```
[{'window': 'boxcar', 'null_to_null_hz': 31.25, 'peak_sidelobe_db': -
13.263647577714902, 'enbw_hz': 15.625}, {'window': 'hann', 'null_to_null_hz':
62.5, 'peak_sidelobe_db': -31.467626478524792, 'enbw_hz': 23.4375},
{'window': 'hamming', 'null_to_null_hz': 62.5, 'peak_sidelobe_db': -
42.6743663692628, 'enbw_hz': 21.294152949245543}, {'window': 'blackman',
'null_to_null_hz': 93.75, 'peak_sidelobe_db': -58.10884368817948, 'enbw_hz':
26.98058390022675}]
```

```
from IPython.display import display, Image
display(Image(filename="results/windows.png"))
```



Overlap and exact FFT convolution

```
print(metrics['overlap'])
```


```
[{'overlap_percent': 0, 'hop': 1024, 'window': 'boxcar', 'frames': 65,
'relative_error': 2.6735228456948584e-16, 'min_denominator': 1.0,
'max_denominator': 1.0, 'elapsed_seconds': 0.04068229999393225},
{'overlap_percent': 0, 'hop': 1024, 'window': 'hann', 'frames': 65,
'expected_failure': 'NOLA failure in retained samples', 'elapsed_seconds':
0.0032158999238163233}, {'overlap_percent': 25.0, 'hop': 768, 'window':
'hann', 'frames': 86, 'relative_error': 3.7200893184236793e-16,
'min_denominator': 0.0428932188134525, 'max_denominator': 1.0,
'elapsed_seconds': 0.004012899938970804}, {'overlap_percent': 50.0, 'hop':
512, 'window': 'hann', 'frames': 128, 'relative_error': 2.444013099020329e-
16, 'min_denominator': 0.5, 'max_denominator': 1.0, 'elapsed_seconds':
0.005510699935257435}, {'overlap_percent': 75.0, 'hop': 256, 'window':
'hann', 'frames': 255, 'relative_error': 2.0087174059560197e-16,
'min_denominator': 1.4999999999999993, 'max_denominator': 1.5000000000000007,
'elapsed_seconds': 0.010108600021339953}, {'overlap_percent': 87.5, 'hop':
128, 'window': 'hann', 'frames': 509, 'relative_error': 1.9368621161930796e-
16, 'min_denominator': 2.999999999999999, 'max_denominator':
3.0000000000000013, 'elapsed_seconds': 0.020057399990037084}]
```

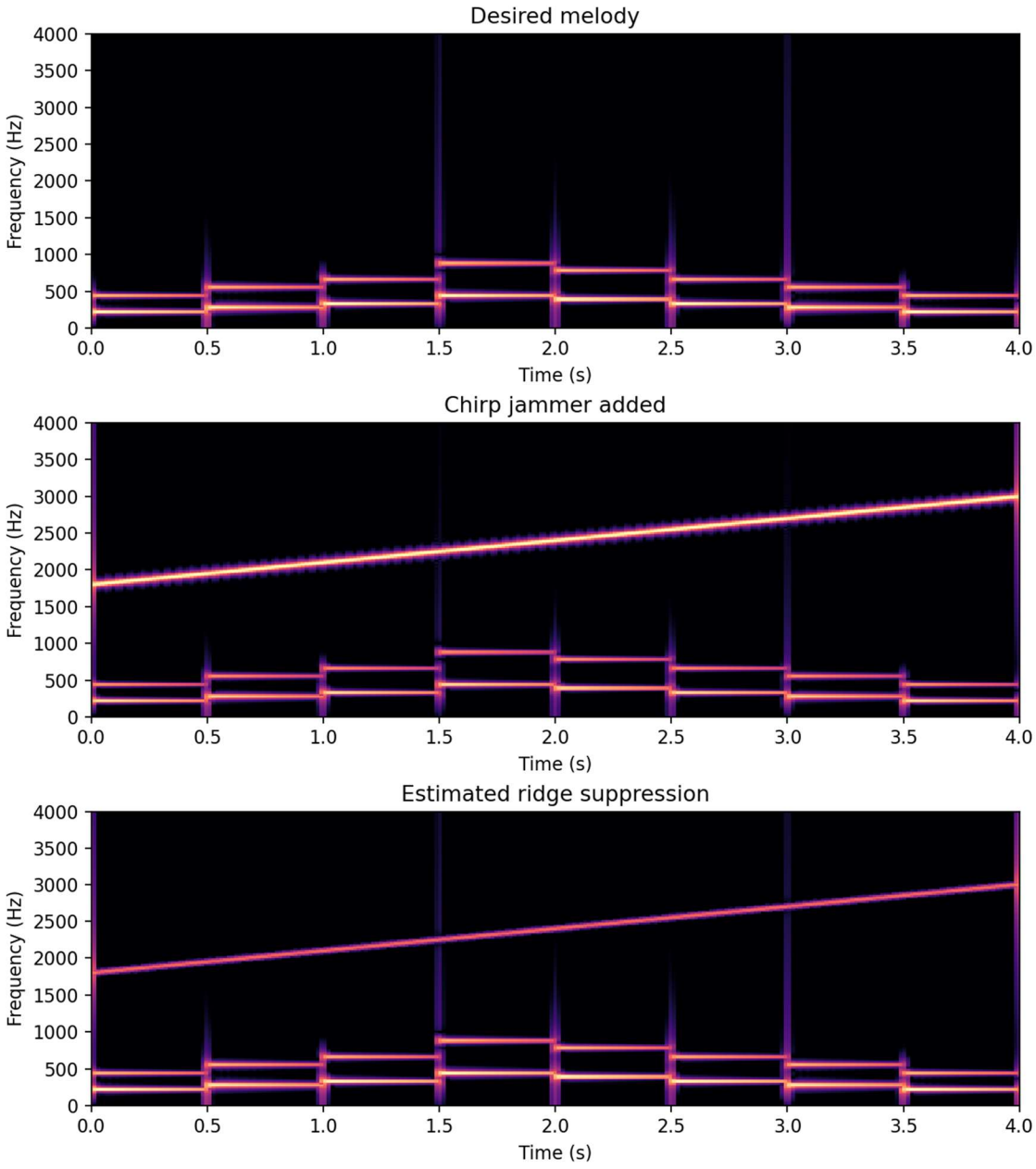

Moving jammer

```
print(metrics['jammer_sweep'])
```


```
[{'L': 256, 'width_hz': 30, 'output_snr_db': -0.8024310924202096,
'jammer_attenuation_db': 8.439462222975777, 'clean_relative_error':
0.0002430896500555O364}, {'L': 256, 'width_hz': 60, 'output_snr_db':
4.843103849910849, 'jammer_attenuation_db': 14.08499339328788,
'clean_relative_error': 0.00037779775091329127}, {'L': 256, 'width_hz': 120,
```

'output_snr_db': 12.481593042684294, 'jammer_attenuation_db': 21.72345862836615, 'clean_relative_error': 0.0005655553072143046}, {'L': 1024, 'width_hz': 30, 'output_snr_db': 11.76507671056524, 'jammer_attenuation_db': 21.007007364925418, 'clean_relative_error': 0.0002810228251903903}, {'L': 1024, 'width_hz': 60, 'output_snr_db': 17.349087100818082, 'jammer_attenuation_db': 26.590936278608414, 'clean_relative_error': 0.000404427480612134}, {'L': 1024, 'width_hz': 120, 'output_snr_db': 19.878447725307534, 'jammer_attenuation_db': 29.120208120571572, 'clean_relative_error': 0.0005753947233051704}, {'L': 4096, 'width_hz': 30, 'output_snr_db': 11.350648144222896, 'jammer_attenuation_db': 20.59250594230059, 'clean_relative_error': 0.00027841472515402395}, {'L': 4096, 'width_hz': 60, 'output_snr_db': 17.0753710187401, 'jammer_attenuation_db': 26.317142591547835, 'clean_relative_error': 0.0004034867182518259}, {'L': 4096, 'width_hz': 120, 'output_snr_db': 19.76949495780763, 'jammer_attenuation_db': 29.01121795144676, 'clean_relative_error': 0.0005751797356772257}]

```
from IPython.display import display, Image
display(Image(filename="results/jammer.png"))
```

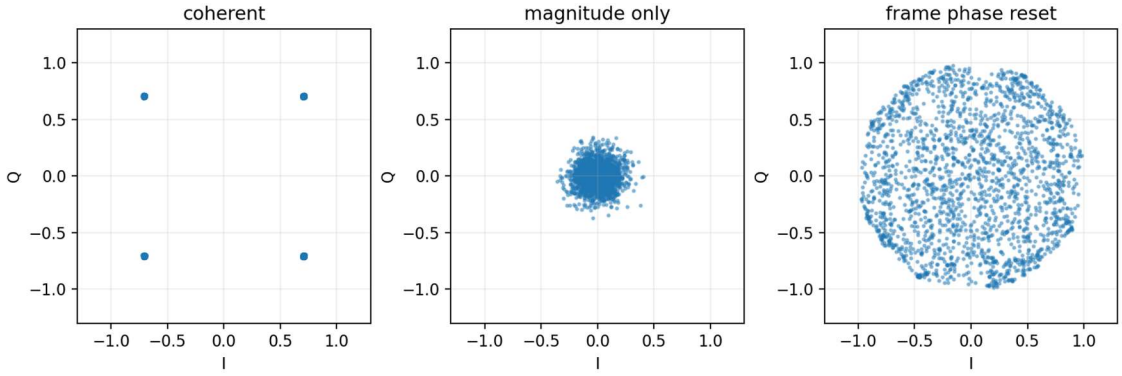


Phase coherent QPSK

```
print(metrics['communications'])
```

```
{'original': {'evm_percent': 0.947661602826916, 'ber': 0.0,
'waveform_relative_error': 0.0}, 'coherent': {'evm_percent':
0.9476616028269153, 'ber': 0.0, 'waveform_relative_error':
2.3268009963511954e-16}, 'magnitude_only': {'evm_percent':
101.02894144350883, 'ber': 0.4990138067061144, 'waveform_relative_error':
1.0103187240492075}, 'frame_phase_reset': {'evm_percent': 127.67102277627777,
'ber': 0.5226824457593688, 'waveform_relative_error': 1.2816860544023}}
```

```
from IPython.display import display, Image
display(Image(filename="results/constellations.png"))
```



Acceptance checks

`main()` checks identity reconstruction, expected Hann NOLA failure, real and complex full FFT convolution, and zero BER after coherent QPSK reassembly. EVM includes finite RRC pulse truncation. WAV round-trip error is separate from float64 identity error. Runtime values vary across machines. See the solution document for the derivations and grading notes.